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Development of a neutron imaging method with the world's highest resolution

- Expected to be used for precise non-destructive testing of products containing hydrogen, lithium, and boron -

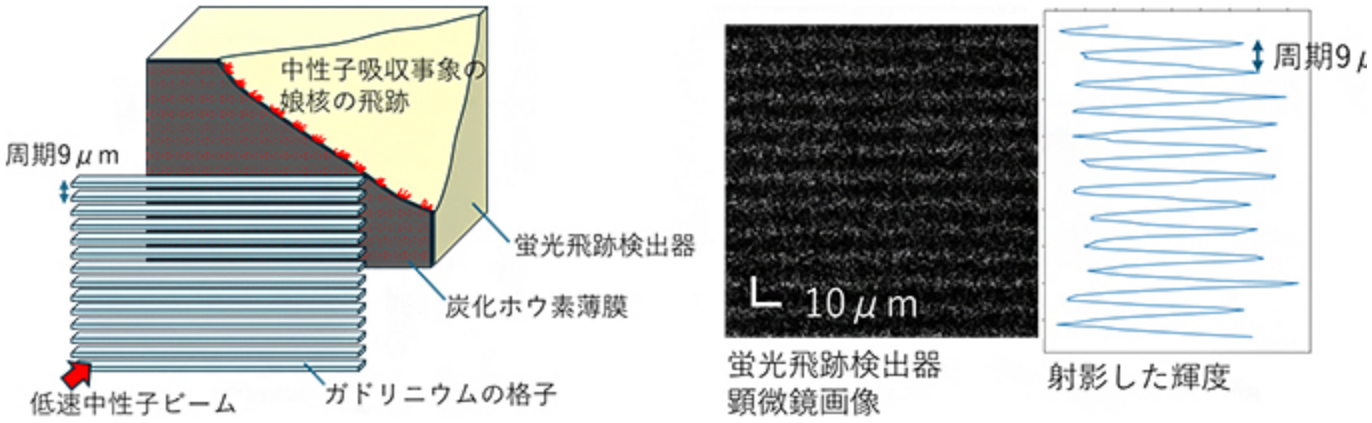
Video text file (PDF 1.3MB)

An international collaborative research group including Takehiko Saito, Principal Investigator at the Saito High Energy Nuclear Science Laboratory, RIKEN Pioneering Research Center, Abdul Muneem, International Program Associate (at the time of the research, currently a Special Researcher at the Neutron Beam Technology Development Team, Center for Advanced Photonics), and Junya Yoshida, Associate Professor at the International Synchrotron Radiation Innovation Smart Research Center, Tohoku University, has developed a neutron imaging method with the world's highest resolution, finer than 1 micrometer (μm, 1 μm is 1/1000 of a millimeter).

The results of this research are expected to contribute to precise non-destructive testing of products containing hydrogen [1] , lithium [2] , and boron [3] , which are difficult to visualize using X-rays .

This time, an international collaborative research group applied high-spatial resolution charged particle tracking detection technology to neutron imaging, and developed a neutron detector that combines a boron carbide thin film with a fluorescent tracking detector. Quantitative evaluation of the resolution of this detector revealed that it had the world's highest accuracy among neutron imaging devices.

This research was published in the online edition of the scientific journal *Scientific Reports (January 24th: January 24th Japan time).*



Schematic diagram of the developed neutron imaging method

background

Nondestructive testing techniques that visualize the interior of objects without destroying them are essential for the development and quality control of modern industrial products. Nondestructive testing of advanced products with fine structures requires micrometer-scale imaging resolution comparable to that of optical microscopes. X-ray transmission imaging (i.e., radiography) is a typical nondestructive testing technique. However, this method observes the shadows formed by X-rays absorbed by electrons in materials, making it difficult to visualize the low-density contrast structures of light elements such as hydrogen, lithium, and boron. These elements are used in products that will support the energy needs of future society, such as hydrogen energy, lithium secondary batteries, and superconducting wires, and therefore require nondestructive testing techniques that are complementary to X-rays.

Neutron beams have therefore attracted attention. Neutrons are one of the particles that make up atomic nuclei, along with protons. They have zero electric charge and decay by themselves with a half-life of about 10 minutes. When neutrons are separated from atomic nuclei to create a slow neutron beam [4] and irradiated onto a material, they are scattered and absorbed relatively well by

certain atomic nuclei, such as hydrogen, lithium-6, and boron-10. Utilizing this property, the spatial distribution of these elements within an object can be visualized by transmission neutron imaging [5] .

There have been several challenges in improving the precision of neutron imaging. Since neutrons have no electric charge and are difficult to detect as they are, they are absorbed by an atomic nucleus that easily absorbs neutrons, and the charged particles (daughter nuclei) emitted during this process are used for detection. In this process, radiation such as gamma and beta rays is generated in an environment where a neutron beam is irradiated, so the detector used must be able to distinguish the neutron signal from the noise of this radiation. Furthermore, neutron beams cannot be refracted or reflected as easily as visible light. Therefore, previous research on neutron imaging used a method that uses a scintillator plate that emits light when a neutron absorption event occurs, optically magnifying the light-emitting spot with a lens and photographing it with a CMOS camera. The best spatial resolution of this method of neutron imaging was approximately 2 μm.

Meanwhile, the international collaborative research group aimed to refine the data using a different approach than previous research.

Research Methods and Results

An international collaborative research group has developed a neutron imaging device that combines a boron carbide thin film with a fluorescent tracking detector. This method applies the high-resolution charged particle tracking detection technology developed by the international collaborative research group for nuclear physics to neutron imaging (Figure 1).

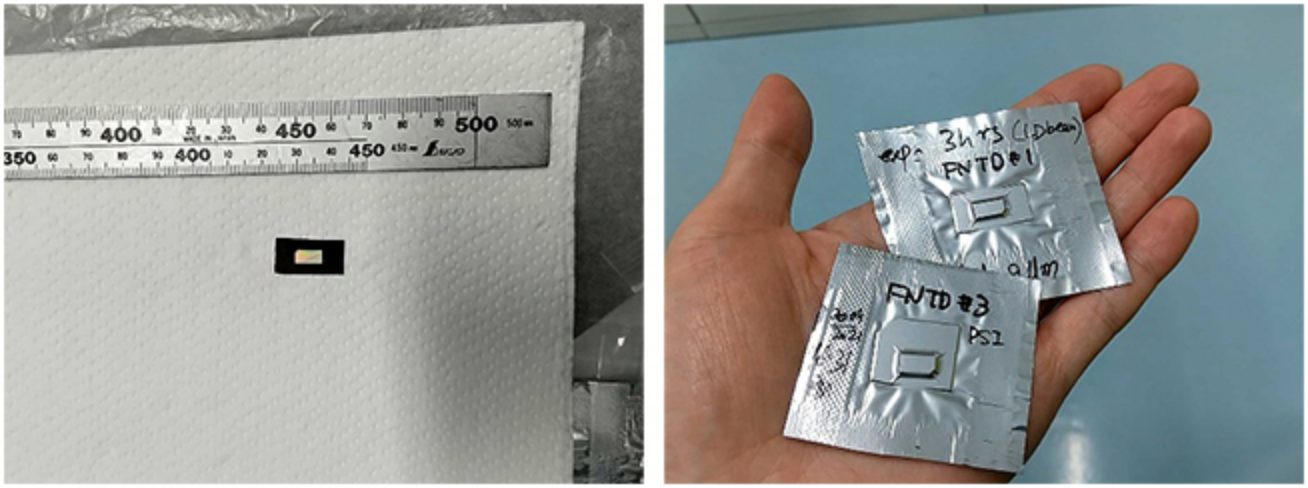


Figure 1 Neutron imaging device combining a boron carbide thin film and a fluorescent tracking detector

left: The black object in the center of the image is a boron carbide thin film formed on the surface of a silicon substrate. The yellow square above it is a fluorescent tracking detector, 4mm long, 8mm wide, and 0.5mm thick.

right: A neutron imaging device was created by tightly contacting a boron carbide thin film and a fluorescent track detector, encasing them in laminate film, and vacuum-packing them. The neutron beam is directed from the back to the front in this photograph. After the experiment, the pack is opened, and the fluorescent track detector is read using a confocal laser microscope.

One of the features of this technique is the use of a thin film of boron carbide (B_4C) in the area where neutrons are converted into daughter nuclei. This film is 230 nanometers (nm, 1 nm is one billionth of a meter) thick and is enriched with 10 boron nuclei. When it absorbs a slow neutron, it releases two daughter nuclei through the reaction " $^{10}B+n \rightarrow {}^{7}Li+\alpha$". These are emitted in opposite directions, and one of them can be detected when it enters the detector.

Another feature of this study is the use of a fluorescent track detector. The fluorescent track detector used in this study is made of a single crystal primarily composed of alumina (Al_2O_3). When a charged particle passes through it, it leaves a trace of its passage. When this detector is illuminated with a red laser light under a microscope called a confocal laser microscope, infrared light is emitted from the point where the charged particle passed. Reading this infrared light

produces an image of the track. For example, by using a 40x magnification objective lens to read an area approximately 106 μm square with a resolution of approximately 1,000 pixels, tracks can be measured with a spatial resolution of approximately 0.1 μm. While this fluorescent track detector has low sensitivity to environmental radiation such as gamma and beta rays, it can visualize each individual daughter nucleus track. While fluorescent track detectors have previously been used as charged particle detectors, this is the first time they have been used in combination with a boron carbide thin film for neutron imaging.

国際共同研究グループは、開発した中性子イメージングデバイスの性能を評価するため、9μmの周期構造を持つガドリニウムの格子[6]を使いました。ガドリニウムには天然元素中で中性子を最もよく吸収するガドリニウム157原子核が含まれており、この元素で作った物体に中性子ビームを当てると高いコントラストで影が生じます。このガドリニウムの格子は入手可能なガドリニウム製の被写体の中で最も微細な構造を持つ物体だったことから、本中性子イメージング手法のイメージング分解能の評価に用いました。実験は茨城県東海村の実験施設J-PARCにある物質・生命科学実験施設で行いました。中性子ビームをガドリニウムの格子に照射し、その隙間を通過してできた縞（しま）状の中性子の模様を、開発した中性イメージングデバイスに記録しました。

図2は、実験に用いた蛍光飛跡検出器を共焦点レーザー顕微鏡下で読み出して得られた画像です。ここでは期待通りにガドリニウム格子によってつくられた周期9μmの縞模様が確認できました。暗い帯状の部分がガドリニウムによって中性子が遮られた部分で、明るい部分が中性子吸収事象から放出された娘核の飛跡が記録されている部分です。

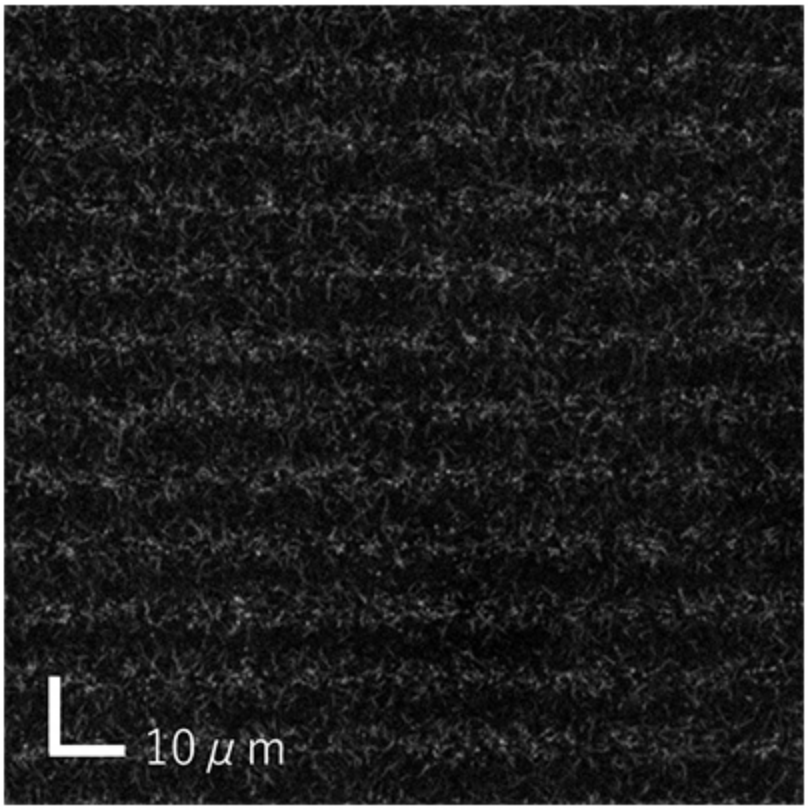


図2 共焦点レーザー顕微鏡で撮影した格子の透過中性子イメージ

ガドリニウム格子を透過した中性子ビームの作った縞模様。明るい帯状の部分は中性子吸収事象から放出された娘核の飛跡が記録されている部分で、暗い部分は中性子が遮られた部分。縞模様の周期はガドリニウム格子と同じ9μm。視野のサイズは約106μm四方。照明光は波長640nmの赤い光で、荷電粒子の影響を受けた箇所から放出される750nmの赤外線を検出して画像化した。

国際共同研究グループは、開発した中性子イメージングデバイスの分解能を評価するため、この縞模様の縁の鮮明さを評価しました（図3）。得られた画像の輝度を、縞と平行な方向に射影すると、山と谷が交互に繰り返されるグラフが得られます。画像の明るい部分がグラフの山に、暗い部分が谷に対応します。この山と谷の間の斜面がどれだけ急であるかを定量的に評価することで、得られた画像の鮮明さの指標としました。ここでは、本手法のイメージング分解能を他の手法と比較できるように以下の2通りの指標を用いました。一つ目の指標は輝度が10％から90％に至るまでの距離で、その平均値は2.27±0.02μmと算出されました。二つ目の指標は画像の縁のぼけ方が正規分布的だとしたときの標準偏差に相当する距離で、0.887±0.009μmと算出されました。

この値は、これまでに報告されていた最も良い空間分解能である約2μmよりも小さく、本中性子イメージングデバイスの空間分解能が世界最高精度であることが示されました。

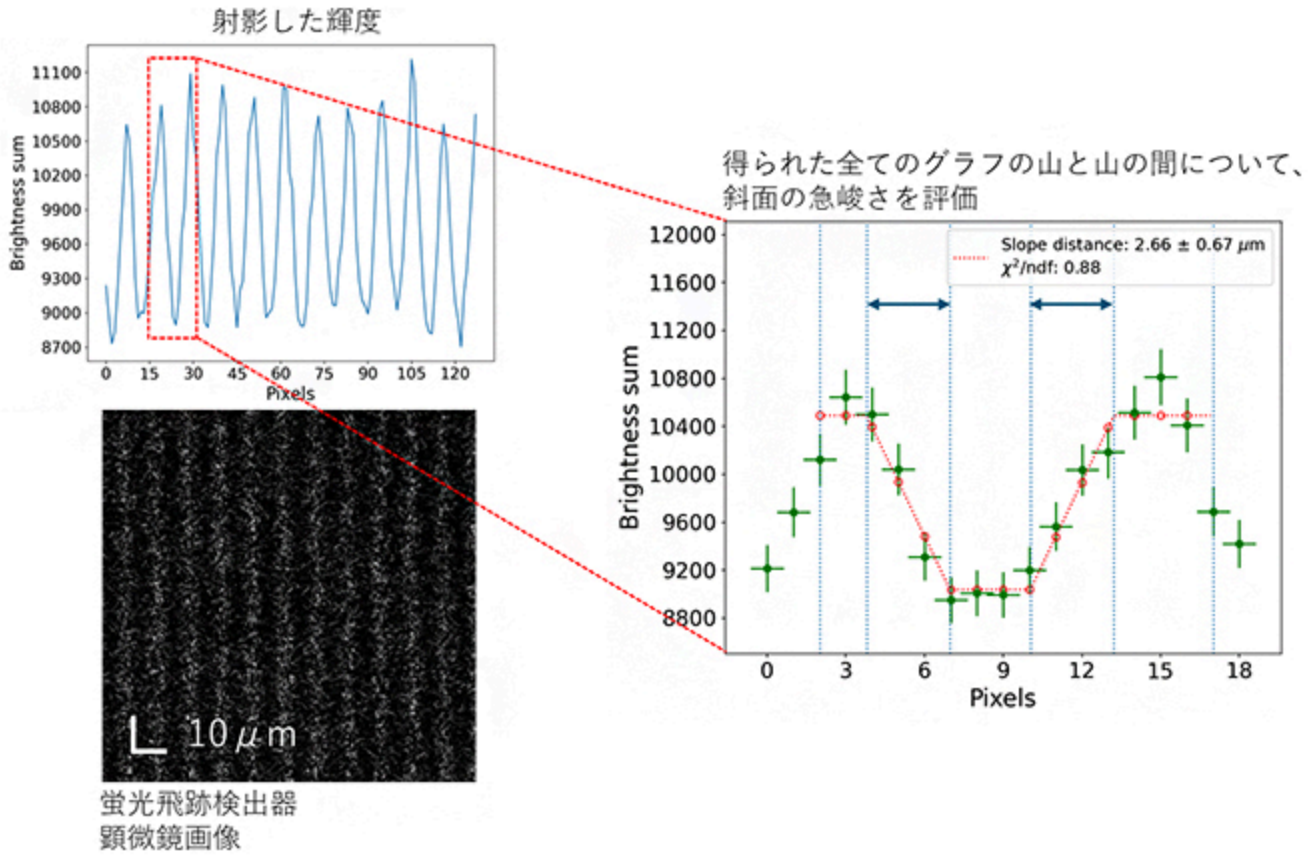


図3 縞模様の縁の鮮明さでイメージング分解能の性能を求めた解析

ガドリニウム格子を透過した中性子ビームの縞模様の画像の輝度を、縞と平行な方向に射影し、山と谷が交互に繰り返されるグラフを得る。画像の明るい部分がグラフの山に、暗い部分が谷に対応する。この山と谷の間の斜面がどれだけ急であるかを定量的に評価した。この画像の赤枠で示した場所では、斜面の距離は2.66±0.67μmと算出された。この解析を得られた543組全ての縞に対して行った上で、イメージング分解能を2通りの指標で算出した。一つ目の指標は斜面の10%から90%の高さに至るまでの距離で、その平均値は2.27±0.02μmと算出された。また二つ目の指標は縞の縁のぼけ方が正規分布的だとしたときの標準偏差を表す値で、0.887±0.009μmと算出された。

今後の期待

今回の研究で、炭化ホウ素薄膜と蛍光飛跡検出器を組み合わせたデバイスによって、1μmを下回る空間分解能の中性子イメージングが実現できることが明らかになりました。国際共同研究グループは今後、この手法の実用化に向けた研究を進めていきます。具体的には、デバイスの大面積化、顕微鏡による読み出しの高速化、どこまで高密度に飛跡を記録できるかの検出器の応答の評価、飛跡の消去[7]法の確立などです。

本中性子イメージングデバイスは、水素、リチウム、ホウ素といった軽元素を含む製品の精密非破壊検査に応用されることが期待されます。これらの元素は、将来の社会のエネルギーを支える重要な製品に用いられている一方、X線による非破壊検査が困難です。そこで今後の開発によって精細な中性子イメージングが実現すれば、非破壊検査を通じてこれらの製品の開発や改良に貢献できると考えられます。

また理研では、RANS（ランズ）[8]と呼ばれる小型中性子源の開発・運用も行っています。国際共同研究グループは、RANSの研究チームとも連携し、将来的に小型中性子源を用いた中性子イメージングへの応用も見据え、中性子源から画像処理までを包括した中性子イメージング技術の開発を行う予定です。

補足説明

1. **水素**
   原子番号1の元素。燃焼させて熱を発生させたり、燃料電池を用いて電気を発生させたり、一時的に余剰になった電気エネルギーの貯蔵に使われたりするなど、将来のエネルギー戦略上の重要な物質として位置付けられている。中性子を散乱しやすい。
2. **リチウム**
   原子番号3の元素。携帯電話やノートPCなどの電源として広く普及しているリチウムイオン二次電池に利用されている。天然のリチウム中に約7.5%含まれるリチウム6原子核が中性子を比較的よく吸収する。
3. **ホウ素**
   原子番号5の元素。39ケルビン（K：絶対温度の単位）という比較的高い温度でも超伝導を示す二ホウ化マグネシウムに使われており、この物質は電力損失のない超伝導ケーブルの素材の候補とされている。天然のホウ素中に約20%含まれるホウ素10原子核が中性子を比較的よく吸収する。

4. **低速の中性子ビーム**

   中性子源から放出される中性子の流れを中性子ビームと呼ぶ。もし中性子源が常温で熱平衡に達している場合、放出される中性子の平均速度は約2,200m毎秒となり、これを熱中性子ビームと呼ぶ。熱中性子ビームを基準として低速の中性子ビームは冷中性子ビーム（平均速度は100m毎秒程度）と呼ぶ。中性子は速度に反比例して原子核に吸収されやすくなるので、透過中性子イメージング（[5]参照）には熱・冷中性子ビームを用いるのが効果的である。

5. **透過中性子イメージング**

   物体に中性子ビームを照射し、透過してきた中性子を検出することで、物体内部にある中性子を吸収しやすい物質の分布を非破壊で可視化する手法。

6. **Gadolinium lattice:**

   Gadolinium, which easily absorbs neutrons, is arranged in a lattice pattern. In this study, gadolinium was arranged with a width of approximately 5 μm and a periodicity of 9 μm. Originally developed for neutron phase imaging, it has the finest structure of any available gadolinium product, so it was used to evaluate the imaging resolution in this study. This gadolinium lattice was fabricated using a precision process called oblique deposition, but its shape is not a perfect rectangle, so the blurring of the obtained image is influenced by the imaging resolution of the detector as well as the bluntness of the corners of the gadolinium lattice.

7. **The fluorescent track detector used in this**

   study can be reused by erasing the tracks with an ultraviolet laser. The international collaborative research group constructed a unique ultraviolet laser irradiation system and demonstrated that even when alpha ray tracks were recorded at a high density of 10,000 lines in an area of 100 μm square, the tracks could be erased with an ultraviolet laser, and that the detector could be reused even after repeating the tracking and erasure cycle at least seven times .

8. **RANS**

   is a compact neutron source system developed by RIKEN. It irradiates a target with a proton beam accelerated by an accelerator, spalling the nucleus and generating a neutron beam. RANS stands for RIKEN accelerator-driven compact neutron systems.

Note)Abdul Muneem, Junya Yoshida, Hiroyuki Ekawa, Masahiro Hino, Katsuya Hirota, Go Ichikawa, Ayumi Kasagi, Masaaki Kitaguchi, Satoshi Kodaira, Kenji Mishima, Jameel-Un Nabi, Manami

Nakagawa, Michio Sakashita, Norihito Saito, Takehiko R. Saito, Satoshi Wada, Nakahiro Yasuda, Study on the reusability of fluorescent nuclear track detectors using optical bleaching, *Radiation Measurements* , Volume 158, 2022

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